Worksheet 2.5

Dominant and recessive

Answers

Red and yellow tomatoes

1. A student conducted a series of experiments with pure breeding red tomatoes and pure breeding yellow tomatoes.

The pollen from the yellow tomatoes was allowed to pollinate the stigmas of the red tomatoes. The seeds were collected and grown to determine the colour of the fruits of the first generation. All the first generation produced red fruit. The first generation plants were allowed to interbreed and the colour of the second generation fruit was determined. The results were:

red fruit = 360 **yellow fruit = 119**

(a) Explain why all the first generation produced red fruit.

The progeny of the first generation are hybrids with a dominant red gene and a recessive yellow gene. Thus all the first generation have red fruit.

(b) Calculate the ratio of red fruit to yellow fruit in the second generation. Round your answer to the nearest whole number.

$\frac{360}{119} = \frac{3}{1}$

(c) Explain why yellow fruit appeared in the second generation.

The first generation are hybrids and there is a 1 in 4 chance that yellow fruits will be formed in the second generation.

(d) Predict the proportion of tomatoes in the second generation that are likely to be pure breeding.

The second generation genotypes are: *RR*, *Rr*, *rR*, *rr*.

The *RR* and *rr* individuals are pure breeding. They will represent half (50%) the second generation.

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Ear lobes

2. The photos below show two types of ear lobes. Free ear lobe (*F*) is dominant to attached ear lobe (*f*).



(a) A homozygous woman with free ear lobes marries a man who has attached ear lobes. Predict the types of lobes their children will inherit.

All will have free lobes as all are hybrids.



Photos: John Wiley & Sons Australia/Photo by Renee Byron

(b) Two parents who are hybrids for free ear lobes have four children. Predict the proportion of free ear lobes to attached ear lobes in their children.

The possible genotypes are:

FF, *Ff*, *fF* and *ff*.

free:attached = 3:1

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